
E-Cell Tech Team Recruitment 2026

Task: Case Competition Slide Showcase

Detailed Task Problem Statement

-  **Role:** Technical Team Associate
-  **Organization:** E-Cell IIT (BHU) Varanasi
-  **Domain:** Full Stack Web Development
-  **Session:** 2026-2027

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1 Introduction and Overview

1.1 Task Objective

The objective of this task is to evaluate your full-stack development skills by building a complete web application. You are required to design and develop a platform to showcase Case Competition slides/PDFs. The application must feature secure authentication, a user-friendly and responsive interface, and proper backend routing for resource management.

2 Core Requirements

2.1 Pages and UI Design

The web application must contain at least two primary pages with a responsive frontend design.

- **Home Page:**
 - Must display a list of all uploaded slides in a paginated format.
 - Should include functionalities for filtering, searching, and sorting the slides.
 - An “Upload” button must be prominently placed on the top right side.
- **Upload Page:**
 - A simple form allowing authenticated users to upload new slides.
 - Must capture necessary metadata: preview image, title, description, and tags.

2.2 Authentication & Security

User authentication is required for modifying data. You must implement JWT-based authentication. This can be achieved through:

- A simple Email and Password login system.
- Or OAuth integration (e.g., Google, GitHub).

3 Technical Stack

You have the flexibility to choose specific technologies within the modern web ecosystem:

- **Frontend:** React or Next.js combined with TailwindCSS for styling.
- **Backend:** Node.js with Express (MERN stack) or similar backend frameworks.
- **Database:** MongoDB, Supabase, Firebase, or any other database of your choice will work.

4 API Routes & Design Plan

The backend must provide a robust RESTful (or similar) API to handle CRUD operations for the slides.

Public vs Protected Routes

Anyone should be able to view the slides without logging in. However, modifying the data (Create, Update, Delete) strictly requires an authenticated session.

4.1 Proposed Routes

- GET /api/slides — (*Public*) Fetch all slides. Supports query parameters for pagination (?page=1&limit=10), filtering, searching, and sorting.
- GET /api/slides/:id — (*Public*) Fetch a single slide by its ID.
- POST /api/slides — (*Protected*) Upload a new slide. Requires JWT token in headers. Body should include title, description, tags, preview image URL/file, and slide URL/file.
- PUT /api/slides/:id — (*Protected*) Update an existing slide's metadata.
- DELETE /api/slides/:id — (*Protected*) Delete a slide.
- POST /api/auth/login — (*Public*) Authenticate user and return a JWT token.
- POST /api/auth/register — (*Public*) Register a new user.

5 Deliverables

Candidates are expected to submit the following:

1. **Readable Code:** Clean code with proper folder structuring.
2. **Documentation:** Adequate comments explaining complex logic.
3. **Deployment:** A live deployed link to the working application. If deployment is not possible, provide a detailed video link demonstrating the application's functionality.
4. **Source Code:** A link to the GitHub repository containing the frontend and backend code.

6 Example Prototype

CaseVault

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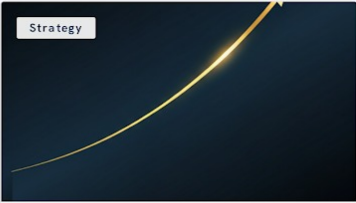
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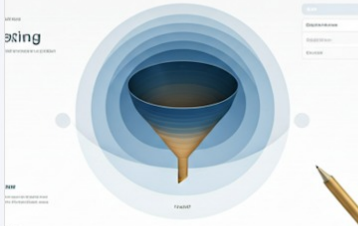


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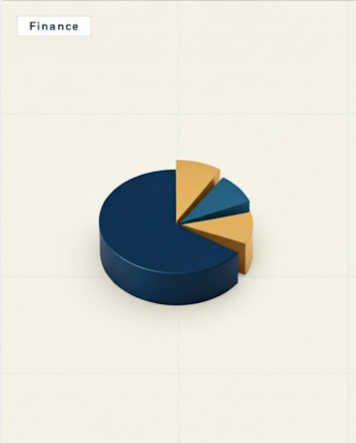


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Figure 1: Home Page

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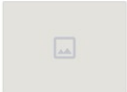
CASE MATERIALS



Drag and drop slides here

or click to browse (.pdf, .pptx, .key)

PREVIEW THUMBNAIL



Choose File

Optimal ratio 16:9. Max size 5MB.

CASE TITLE

Enter a prestigious title...

COMPETITION NAME

e.g., Global Strategy Case 2024

YEAR

2024

CATEGORY

Strategy

EXECUTIVE SUMMARY

Provide a concise summary of the case problem and your strategic solution...

Save Draft

Publish to Vault

Figure 2: Upload Page

7 Deadline and Submission Link

Submission Details

- **Deadline:** 19 June 2026, End of Day (EOD)
- **Submission Link:** <https://forms.gle/StSGetKwqVQbJzQu6>